Cellular–LEO Interference Coordination

Coherent Beacons and Blind Spatial Sensing

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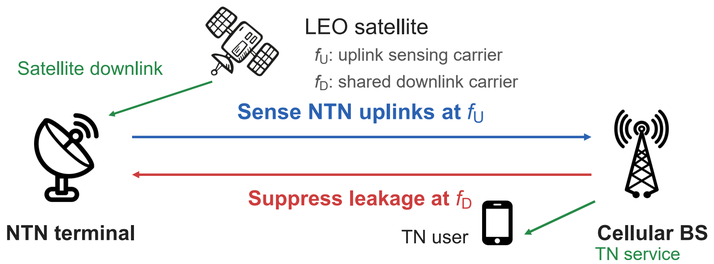
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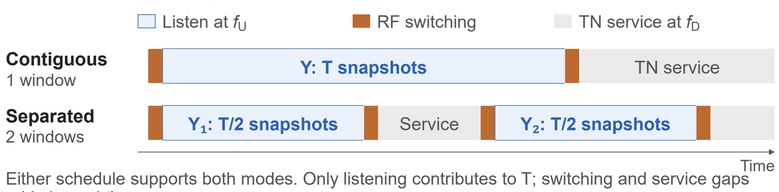
1 SENSING & COORDINATION

Share spectrum while protecting NTN downlink receivers

Sense nearby NTN uplinks, then shape cellular beams, protecting NTN downlink receivers while serving TN users.



Shared-RF listening: one or several windows



Either schedule supports both modes. Only listening contributes to T; switching and service gaps add elapsed time.

A COHERENT

Known beacon + matched filtering
Estimate the reciprocal channel.
Uses a timing / phase reference.

$$\widehat{\mathbf{h}}_k = \mathbf{h}_k + \mathbf{v}_k$$

B NONCOHERENT

Unknown uplink + covariance / MUSIC
Pool windows; estimate angles/powers.
No inter-window phase lock needed.

$$\widehat{\mathbf{R}} = T^{-1}\sum_{\ell} \mathbf{Y}_\ell \mathbf{Y}_\ell^H$$

Beamforming design with adjustable protection

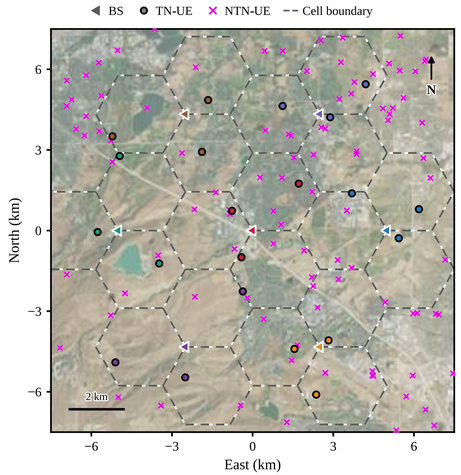
$$\widehat{\mathbf{w}} = \mathbf{v}_{\max}\left(\mathbf{u}_0\mathbf{u}_0^H - \lambda \mathbf{B}\right), \quad \|\widehat{\mathbf{w}}\| = 1$$

$$\mathbf{B}_{\text{coh}} = \sum_k \widehat{\mathbf{h}}_k \widehat{\mathbf{h}}_k^H \qquad \mathbf{B}_{\text{nc}} = \sum_k \widehat{G}_k^{D,\,\text{nom}} \widehat{\mathbf{u}}_k^D (\widehat{\mathbf{u}}_k^D)^H$$

u_0: TN-channel direction. λ balances gain and leakage. Rebuild noncoherent directions at the DL carrier; nominal weights are not a hard INR guarantee.

2 SPATIAL INTERFERENCE CONTROL

Suburban area, Colorado • 7 sites / 21 sectors



SITE-SPECIFIC STUDY

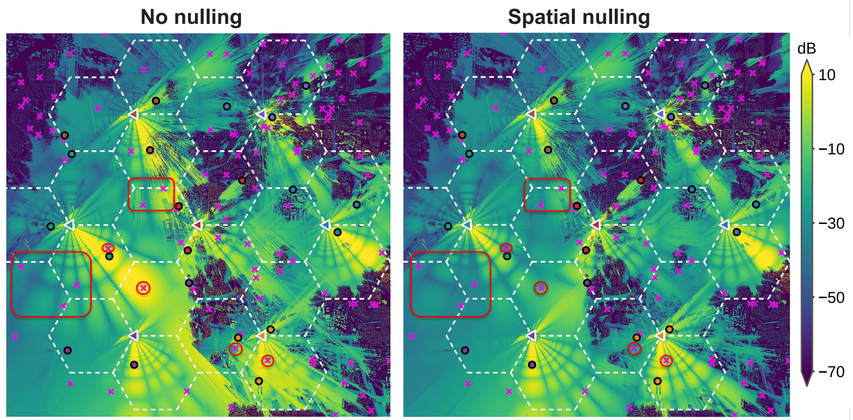
15 × 15 km
Aerial context with ray-traced propagation

100 NTN terminals
50 rooftop + 50 outdoor

8 × 8 BS arrays
f_c = 10 GHz · 200 MHz
35 dBm per sector

Nulling criteria
λ = 10^{12}
Depth-3 multipath evaluation

Aggregate NTN INR: before and after nulling



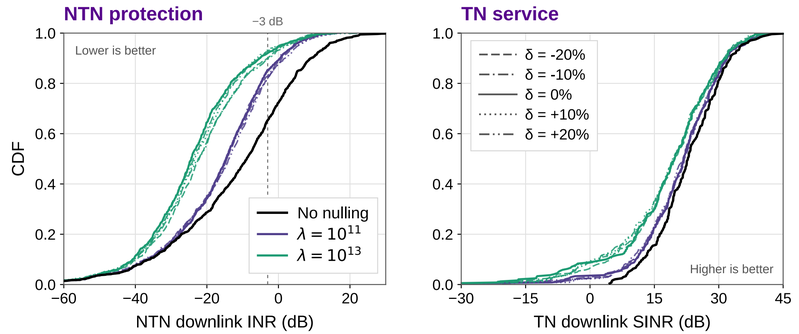
Multipath beam design and evaluation

3 NONCOHERENT PERFORMANCE

Noncoherent sensing at 6.4–9.6 GHz; DL at 8 GHz

4 sites / 12 sectors · 20 paired drops · 8 × 8 arrays
918 linked NTN / 240 TN samples; MDL-MUSIC.

δ = (f_U − f_D) / f_D; finite-snapshot error excluded



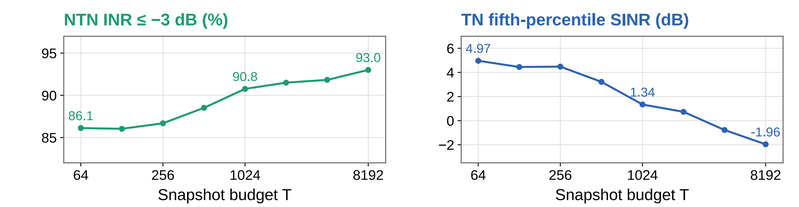
65.3% → 85.2% NTN protection; 1.05 dB TN median SINR loss

NTN INR ≤ −3 dB · δ = 0 · λ = 10^{11}

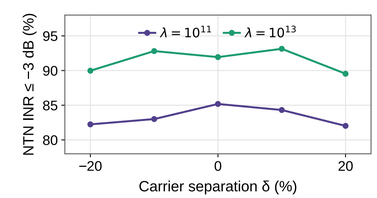
At δ = 0	Baseline	λ = 10^{11}	λ = 10^{13}
NTN fraction ≤ −3 dB	65.3%	85.2%	91.9%
TN median SINR (dB)	22.82	21.77	19.76
TN fifth-percentile SINR (dB)	9.37	6.96	−7.00

Finite snapshots: stronger protection, higher TN cost

8 GHz UL/DL · λ = 10^{13} · 20 deployments × 5 repeats

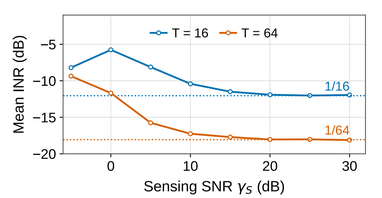


NTN protection across UL–DL carrier gaps



89.5–93.1% meet INR ≤ −3 dB
λ = 10^{13} · tested carrier gaps from −20% to +20%

More snapshots → lower INR plateau



4× snapshots → 6 dB lower plateau
γI = γS · T = 16 → 64 · N = 8, |ρ| = 0.8
Single-victim benchmark; 6,000 trials/point.

4 WHAT LIMITS THE NULL DEPTH?

Single-victim hard null · fixed directions and energies · white Gaussian noise

Coherent: sensing SNR → residual INR

LOW SENSING SNR · exact first-order coefficient

$$\mathbb{E}[\text{INR}] \sim \frac{\mathcal{E}_D}{\mathcal{E}_U}\gamma_U\left[(N-1)r + \frac{1-r}{N-1}\right]$$

HIGH SENSING SNR: LIMIT

$$\mathbb{E}[\text{INR}] \to \frac{\mathcal{E}_D}{\mathcal{E}_U}$$

ALL SENSING SNR: BOUND

$$\mathbb{E}[\text{INR}] \le N\frac{\mathcal{E}_D}{\mathcal{E}_U}$$

$\mathcal{E}_D$: Downlink energy per symbol
$\mathcal{E}_U$: Total uplink training energy
γ_U: Sensing SNR per antenna; N antennas; r = |ρ|².

Noncoherent: sensing SNR → residual INR

LOW SENSING SNR · large-array, small-r approximation

$$\eta_{\text{LM}} \simeq 0 \;\Rightarrow\; \mathbb{E}[\text{INR}] \approx \gamma_I r$$

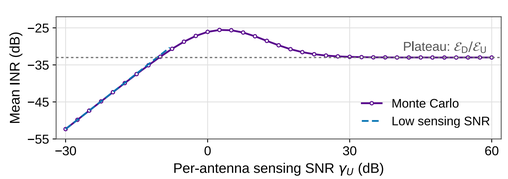
HIGH SENSING SNR · common signature; fixed N, T

$$\mathbb{E}[|\widehat{\mathbf{w}}^H \mathbf{u}_1|^2] \sim (T\gamma_S)^{-1}$$

$$\gamma_I = c\gamma_S \;\Rightarrow\; \mathbb{E}[\text{INR}] \to c/T$$

T: snapshots; γS: array sensing SNR; γI: raw INR.
η: direction overlap; fixed |ρ| < 1 in both modes.

A plateau is not an INR ceiling



$N = 16$, $r = 0.779$, $\mathcal{E}_D/\mathcal{E}_U = 1/2000$
Channel-gain sweep at fixed energies; 500,000 trials/point.

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